

Karish Grover

✉ karishg@cs.cmu.edu

🌐 <https://karish-grover.github.io>

in karishgrover

✉ @karishgrover

🐙 karish-grover

🔍 Google Scholar

Education

- 📖 **Carnegie Mellon University** Pittsburgh, PA
Ph.D., Machine Learning 2023 – 2028
- ◇ Advisors: **Prof. Christos Faloutsos** and **Prof. Geoffrey J. Gordon**.
 - ◇ **Amazon AI Ph.D. Fellowship** (a.y. 2025–2027)
 - ◇ **Meta AIM Ph.D. Fellowship** (a.y. 2025–2026)
- 📖 **Indraprastha Institute of Information Technology, Delhi** Delhi, India
B.Tech., Computer Science and Artificial Intelligence | GPA: 9.2/10.0 2019 – 2023
- ◇ Best **Undergraduate Thesis Award**.
 - ◇ **Chanakya Undergraduate Fellowship** (a.y. 2022–2023)

Refereed Publications

- [A1] CurvGAD: Leveraging Curvature for Enhanced Graph Anomaly Detection  
K. Grover, G. J. Gordon, and C. Faloutsos
42nd International Conference on Machine Learning [ICML 25]
- [A2] Spectro-Riemannian Graph Neural Networks  
K. Grover, H. Yu, X. Song, Q. Zhu, H. Xie, V. N. Ioannidis, and C. Faloutsos
13th International Conference on Learning Representations [ICLR 25]
- [A3] Discourse-Aware Hyperbolic Fourier Co-Attention for Social-Text Classification  
K. Grover, S. M. P. Angara, M. S. Akhtar, and T. Chakraborty
36th International Conference on Neural Information Processing Systems (Oral Paper) [NeurIPS 22]

Patents

- [B1] Dependency Aware Incident Linking in Large Cloud Systems 
S. Ghosh, J. Wong, C. Bansal, R. Namineni, M. Verma, S. Rajmohan, and **K. Grover**
US Patent App. 18/520,931 [US Patent 24]

Other Publications

- [C1] Dependency Aware Incident Linking in Large Cloud Systems 
S. Ghosh, **K. Grover**, J. Wong, C. Bansal, R. Namineni, M. Verma, and S. Rajmohan
ACM Web Conference 2024 [WWW 24]
- [C2] Multi-Relational Graph Transformer for Automatic Short Answer Grading  
R. Agarwal, V. Khurana*, **K. Grover***, M. Mohania, and V. Goyal
Conference of the North American Chapter of the Association for Computational Linguistics [NAACL 22]

Work Experience

- 📖 **Meta, Visiting Researcher** Pittsburgh, PA
*Advised by: **Dr. Yinglong Xia** and **Dr. Hanqing Zeng*** Sep 2025 – Present
- ◇ *Team:* Meta AI – Modern Recommendation Systems (MRS).
 - ◇ *Description:* Graph ML and geometric representation learning for recommender systems; Mixed-curvature (geometric) generative recommender systems; Curvature-based language foundation models.

Work Experience (continued)

- **Amazon, Applied Scientist Intern** Santa Clara, CA
Advised by: **Dr. Theodore Vasiloudis** and **Dr. Xiang Song** May 2025 – Aug 2025
- ◇ **Team:** Amazon Web Services (AWS) Deep Engine Science — Graph ML (AGML).
 - ◇ **Description:** Multi-task self-supervised learning and control-theoretic scheduling on graphs; We introduce Control-G, the *first* control-theoretic approach to multi-objective graph SSL, which uses a task difficulty-aware Proportional-Integral-Derivative (PID) scheduler to decide which task to train next.
- **Amazon, Applied Scientist Intern** Santa Clara, CA
Advised by: **Dr. Xiang Song**, **Dr. Vassilis Ioannidis** and **Dr. Qi Zhu** May 2024 – Aug 2024
- ◇ **Team:** Amazon Web Services (AWS) Deep Engine Science — Graph ML (AGML).
 - ◇ **Project [A2]:** Geometric and spectral graph representation learning; We propose Spectro-Riemannian Graph Neural Networks (CUSP) - a mixed-curvature spectral graph neural network that learns spectral filters to optimize node embeddings in products of constant curvature manifolds.
- **Microsoft, Research Intern** Bengaluru, India
Advised by: **Dr. Xuchao Zhang** and **Dr. Supriyo Ghosh** Jan 2023 – Jul 2023
- ◇ **Team:** Microsoft Research × M365 Research AIOps.
 - ◇ **Project [C1], [B1]:** AIOps and incident management for large-scale cloud services; We propose a dependency-aware incident linking (DiLink) framework which leverages both textual and service dependency graph information to improve the accuracy and coverage of incident links.
- **LinkedIn, Applied Research Intern** Bengaluru, India
Manager: **S. M. Phaneendra Angara** May 2022 – Aug 2022
- ◇ **Team:** LinkedIn AI — Misinformation & Virality.
 - ◇ **Project [A3]:** Geometric representation learning for misinformation and virality; hyperbolic kernel annular attention with RL-based propagation-subgraph selection for explainable fake-news detection on social platforms.

Awards and Achievements

- Oct 2025 ■ **Amazon AI Ph.D. Fellowship** — honored to receive the Amazon fellowship (a.y. 2025-2027) supporting my Ph.D. research at CMU. [
- Sept 2025 ■ **Meta AIM Ph.D. Fellowship** — received the Meta fellowship (a.y. 2025-2026) to support my Ph.D. research, as part of the **Meta AI Mentorship program**. [
- Oct 2023 ■ **Best Undergraduate Thesis Award**, IIIT-Delhi — university-wide award (gold medal) for UG thesis on geometric deep learning for graphs and language [A3]. [
- Sept 2022 ■ **Kartikeya Gupta Memorial Scholarship**, IIIT-Delhi — single university-wide awardee for outstanding all-round academic and co-curricular performance. [
- Aug 2022 ■ **Dean's Award for Academic Excellence**, IIIT-Delhi — recognition for top-tier academic performance for three consecutive years (a.y. 2021-2023). [
- Nov 2021 ■ **Chanakya UG Fellowship**, national cohort (20 awardees) — Received fellowship (a.y. 2021-2022) to support my undergraduate research at IIITD. Awarded by **iHub-Anubhuti**. [
- Jul 2021 ■ **First Position, HAH@IberLEF 2021 (SEPLN) shared task** — winning system for humorous tweet classification at **HAHA subtask** at IberLEF conference. [
- Sep 2019 ■ **Atal Tinkering Grant**, Intel & NITI Aayog — \$20,000 grant to establish a Tinkering Lab; built an assistive navigation robot for visually impaired users. [